

Google Cloud Generative AI Fundamentals

This course guides you through a curated collection of content on generative AI products and technologies, from the fundamentals of Large Language Models to how to create and deploy generative AI solutions on Google Cloud.

DURATION*

8h

LEVEL

Beginner

FEE

HK\$10,000

What you'll learn?

- Learn fundamental concepts of generative AI including how it works, major model types, and applications like content creation and task automation.
- Work with key generative AI techniques like large language models, training data, prompts, encoder-decoder models, and attention to build text generation and image captioning models.
- Implement responsible AI principles and practices to positively impact development and deployment of generative models.
- Leverage Google tools like Generative AI Studio, Vertex AI, and pre-trained models like BERT to develop, deploy, and monitor generative AI applications.
- Gain hands-on experience building end-to-end generative AI solutions from data preprocessing to training models to deployment for text classification, text generation, and image captioning.

Who is this course for?

- Beginners in AI/ML looking to skill up on generative modeling and deploy practical solutions.

*Please refer to the latest schedule on the website of [RTTP \(Registered Public Courses\)](#)

Module 1

Objectives

Introduction to Generative AI

- Learn fundamental concepts like supervised vs unsupervised learning, discriminative vs generative models, and how generative AI works.
 - Understand the major generative AI model types like text-to-text, text-to-image, text-to-video, and foundation models.
 - See examples of generative AI applications like code generation, content creation, task automation, and conversational agents.
 - Discover Google tools to build generative AI applications without coding like Generative AI Studio, Generative AI App Builder, and PaLM API.
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Module 2

Objectives

Introduction to Large Language Models

- Learn what generative AI is and how it works, including key concepts like large language models, training data, and prompts.
 - Understand the major types of generative AI models like text-to-text, text-to-image, text-to-video, and foundation models.
 - Learn how to control generative model outputs through careful prompt design.
 - Discover generative AI applications like creative content generation, task automation, and code generation.
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Module 3

Objectives

Introduction to Responsible AI

- Understand why Google has put AI principles in place.
- Identify the need for a responsible AI practice within an organization.

- Recognize that decisions made at all stages of a project have an impact on responsible AI.
 - Recognize that organizations can design AI to fit their own business needs and values.
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Module 4

Generative AI Fundamentals

Objectives

- Earn a skill badge by completing the Introduction to Generative AI, Introduction to Large Language Models and Introduction to Responsible AI courses. By passing the final quiz, you'll demonstrate your understanding of foundational concepts in generative AI skill badge
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Module 5

Introduction to Image Generation

Objectives

- Introduce diffusion models
 - Understand the theory behind diffusion models
 - Master how to train and deploy diffusion models on Vertex AI.
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Module 6

Encoder-Decoder Architecture

Objectives

- Learn how to implement a character-based text generator using an encoder-decoder neural network architecture with TensorFlow Keras.
- Understand how to preprocess text data and create input and target sequences for the neural network.

- Learn how temperature controls randomness vs determinism in decoding model outputs.
 - See an example of generating creative text content like poetry using a trained encoder-decoder model.
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Module 7

Attention Mechanism

Objectives

- Learn how attention works
 - How attention can be used to improve the performance of a variety of machine learning tasks, including machine translation, text summarization, and question answering.
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Module 8

Transformer Models and BERT Model

Objectives

- Learn how to load a pre-trained BERT NLP model from TensorFlow Hub and use it for text classification.
 - Understand how to preprocess text data and fine-tune BERT for a downstream classification task like sentiment analysis.
 - See how to export a trained BERT model and deploy it on Vertex AI to serve predictions through an online endpoint.
 - Build an end-to-end workflow from loading BERT and training to deploying a text classification model on Vertex AI.
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Module 9

Create Image Captioning Models

Objectives

- Learn how to build an image captioning model by combining a CNN image encoder and RNN text decoder with attention.
 - Understand how to preprocess image and text data, create training sequences, and tokenize text for the model.
 - See how to train the encoder-decoder model end-to-end and generate captions by feeding images to the trained model.
 - Learn how attention helps the decoder focus on relevant parts of the image when generating each word of the caption.
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Module 10**Introduction to Generative AI Studio****Objectives**

- Learn what Generative AI and Generative AI Studio are and how they allow you to generate content like text, images, and audio.
- Understand the core capabilities of Generative AI Studio including prompt design, conversation creation, and model tuning.
- See examples of using Generative AI Studio to design prompts, define conversation contexts, and launch tuning jobs to customize language models.
- Learn how Generative AI Studio provides an easy way to prototype and deploy generative AI models without coding through its graphical interface.